

Foundations of Robotics

Complete student lesson for course code **2371**.

Revision 2021 Semester 2 Program Core 4 modules

Course snapshot

Scheme: 3 (L:3, T:0, P:0)

Credits: 3

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

Course Dashboard

Course code

2371

Semester

2

Credits

3

Teaching scheme

3 (L:3, T:0, P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Fundamentals and Applications of Robots	13	CO1
2	Autonomous Robots: Line Follower and Obstacle Avoider	15	CO2
3	Sensors in Robotic Applications	15	CO3
4	Actuators, Drives and Sensor Selection	15	CO4

Prerequisites

- Basic principles and theorems of Engineering Physics and Chemistry — Physics and Chemistry, Semester 1 & 2.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. To explore the broad scope of robotic applications.
2. To introduce the functional elements of Robotics.
3. To understand the basic concepts associated with the design, functioning, applications and social aspects of robots.
4. To study about the electrical drive systems and sensors used in robotics for various applications.
5. To develop basic robot construction skills.

Course Outcomes

1. CO1: Understand the broad concepts of robotics and its applications. (13 hours; Understanding)
2. CO2: Develop basic kind of autonomous robots. (15 hours; Applying)
3. CO3: Select and use sensors in various robotic application contexts. (15 hours; Applying)
4. CO4: Select and use actuators in various robotic application contexts. (15 hours; Applying)

CO-PO mapping as supplied

CO1: PO1 = 3.

CO2: PO2 = 3.

CO3: PO1 = 3.

CO4: PO1 = 3.

Legend supplied in PDF: 3 = Strongly mapped, 2 = Moderately mapped, 1 = Weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Fundamentals, types and need of robots	1

Module	Topic code	Topic	Hours
module-1	M1.02	Characteristics of robots	3
module-1	M1.03	Applications of robots	3
module-1	M1.04	Industrial applications	4
module-1	M1.05	Parts and functions of robots	2
module-2	M2.01	Basic principle of a line follower	2
module-2	M2.02	Assembling sensors	2
module-2	M2.03	Assembling the line follower	2
module-2	M2.04	Interfacing the microcontroller with motor driver and sensor IC	3
module-2	M2.05	Basic principle of moving	3
module-2	M2.06	Assembling the obstacle avoider	3
module-2	M2.ST1	Series Test I	1
module-3	M3.01	Basic idea of sensors	1
module-3	M3.02	Internal sensors: position and velocity sensors	3
module-3	M3.03	External sensors: proximity, force and torque	3
module-3	M3.04	Sensing variables and sensors for robotics	4
module-3	M3.05	Variable-recognition systems	4
module-4	M4.01	Building blocks of a robot	2
module-4	M4.02	Actuators and DC motors	3
module-4	M4.03	Speed and direction control of drives	3
module-4	M4.04	Nontraditional sensors and actuators	2
module-4	M4.05	Optical, inertial, thermal, chemical, biosensor and other common sensors	3
module-4	M4.06	Selection of sensors and actuators for self-driving cars	2
module-4	M4.ST2	Series Test II	1

Learning Roadmap

1. Fundamentals and Applications of Robots
CO1 · 13 hours

2. Autonomous Robots: Line Follower and Obstacle Avoider
CO2 · 15 hours

3. Sensors in Robotic Applications
CO3 · 15 hours

4. Actuators, Drives and Sensor Selection
CO4 · 15 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Fundamentals and Applications of Robots

Robotics combines mechanical structure, sensing, actuation, control, computation, and purposeful interaction with the environment. This module establishes the vocabulary needed to understand why robots are used, how they are classified, what characteristics matter, and where different applications place different demands on a robot.

Hours: 13

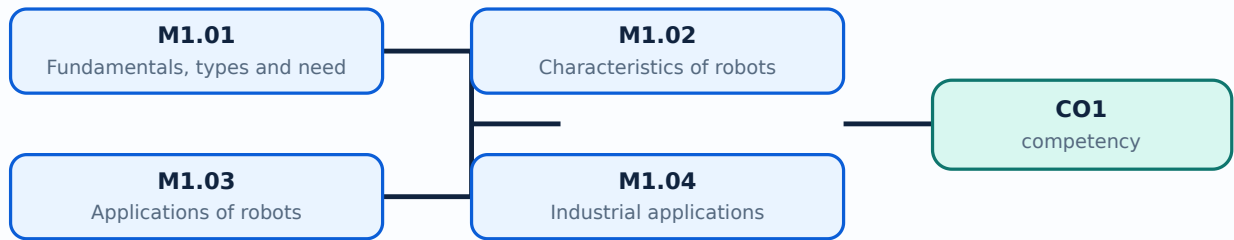
CO1 · Bloom level: Understanding

Learning goals

- Define robot and robotics and explain the need for robots.
- Classify robots and describe their characteristics, advantages, and disadvantages.
- Identify major robot parts and connect them with medical, mining, space, defence, security, domestic, entertainment, and industrial applications.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.01 — Fundamentals, types and need of robots (1 hours)

Concept and explanation

A robot is a programmable machine capable of carrying out actions through a combination of structure, actuation, sensing, and control. Robotics is the field concerned with the design, construction, operation, and application of robots. The need for robots arises when tasks are repetitive, hazardous, inaccessible, highly precise, or require sustained operation. Types may be classified by application, mobility, control, mechanical configuration, or level of autonomy; the classification should state the basis used.

Malayalam support: Robot ഫ്രോം പ്രോഗ്രാമബിൾ മെഷീൻ; sensing, actuation, structure, control ഫ്രോം ഫ്രോം ഫ്രോം. Hazardous, repetitive, inaccessible, precise tasks-ഫ്രോം robots ഫ്രോം.

Study points

- Always state the classification basis; separate a robot from an ordinary automated machine by discussing programmability and interaction; identify the task before selecting a robot type.

Engineering / workplace application: Material handling, inspection, surgery assistance, exploration, and hazardous-environment work illustrate why robotic systems are useful.

Illustrative example: For a warehouse picking task, list the required mobility, payload, sensing, and repeatability before choosing a robot category.

Common mistake: Calling every automatic device a robot without considering programmability, sensing, and task-oriented action leads to an imprecise definition.

— M1.02 — Characteristics of robots (3 hours)

Concept and explanation

Important characteristics include programmability, repeatability, accuracy, payload, reach or workspace, degrees of freedom, speed, resolution, reliability, and the ability to sense or respond to conditions. No single characteristic is universally best: a high-speed robot may not be the most accurate, and a large workspace may require a heavier structure. Characteristics must be related to the application and specified conditions.

Malayalam support: Repeatability, accuracy, payload, workspace, degrees of freedom, speed, resolution റോബട്ട് robot-റോബട്ട് characteristics സ്വഭാവം. Application പ്രയോഗം priority പ്രാധാന്യം.

Study points

- Define each characteristic before comparing systems; distinguish accuracy from repeatability; record units and operating conditions; relate a specification to a task requirement.

Engineering / workplace application: Welding requires repeatable path control; a mobile inspection robot may prioritise battery life, sensing, and obstacle handling.

Illustrative example: Compare two hypothetical robots for pick-and-place: one with higher speed and one with higher repeatability. Explain which characteristic the task needs first.

Common mistake: Accuracy means closeness to the desired value, whereas repeatability means consistency of repeated results; treating them as identical is a common error.

– M1.03 — Applications of robots (3 hours)

Concept and explanation

The syllabus lists medical, mining, space, defence, security, domestic, entertainment, and industrial applications. Applications differ in environment, human interaction, payload, sensing, autonomy, safety, and reliability. A useful application analysis identifies the task, environment, human risk, required sensors, actuator demands, and the reason a robot is preferable to direct human operation.

Malayalam support: Medical, mining, space, defence, security, domestic, entertainment, industrial അപ്ലിക്കേഷനുകൾ applications പട്ടിക. Task, environment, risk, sensing, actuation പട്ടികകൾ application analyse പട്ടിക.

Study points

- Use a structured application table; separate current examples from future possibilities; mention safety and human interaction; do not claim autonomy without describing sensing and control.

Engineering / workplace application: Mining and space robots reduce human exposure to dangerous or inaccessible environments; industrial robots support material handling, welding, spray painting, and machining.

Illustrative example: For a space inspection robot, identify the communication delay, low-maintenance requirement, visual sensing, and fault-tolerant behaviour that influence the design.

Common mistake: Listing application names without explaining the task or engineering reason does not demonstrate understanding.

– M1.04 — Industrial applications (4 hours)

Concept and explanation

Industrial robot applications in the syllabus include material handling, welding, spray painting, and machining. Material handling depends on payload, reach, gripper design, and cycle time. Welding needs path accuracy and process coordination. Spray painting needs controlled motion, coverage, and safe handling of vapour or overspray. Machining needs stiffness, tool control, and suitable force management. Application-specific tooling is often as important as the robot arm.

Malayalam support: Material handling- payload/reach/gripper പ്ലാറ്റ്ഫോം; welding- path accuracy; spray painting- coverage/safety; machining- stiffness/tool control പ്ലാറ്റ്ഫോം.

Study points

- Describe the end effector and process constraint; check workspace and collision risk; distinguish repeatability from surface quality; include guarding and safe operation.

Engineering / workplace application: Robotic cells improve repeatability and reduce exposure to fumes, heat, sharp edges, and repetitive strain when integrated with appropriate guarding.

Illustrative example: Choose a suitable end effector for a welding task and a different one for material handling. State the reason for each choice.

Common mistake: Installing a robot without analysing fixtures, tooling, workspace, and safety interlocks can create a dangerous but apparently automated cell.

— M1.05 — Parts and functions of robots (2 hours)

Concept and explanation

A robot system commonly includes a manipulator or mechanical structure, joints and links, actuators, drives, sensors, controller, power supply, end effector, and user/programming interface. The mechanical structure positions the tool, actuators produce motion, drives regulate actuator power, sensors provide state or environment information, and the controller executes the programmed decision or trajectory. The robot must be considered as a system rather than an isolated arm.

Malayalam support: Manipulator/link/joint, actuator, drive, sensor, controller, power supply, end effector എന്നിവ റോബറ്റ് സിസ്റ്റം-യുടെ ഭാഗങ്ങൾ. ഏതെങ്കിലും ഭാഗം-സിസ്റ്റം-യെ സംബന്ധിച്ചതിന്റെ പ്രവർത്തനം വിശദീകരിക്കുക.

Study points

- Draw a block diagram with signal and power paths; name the end effector separately from the arm; explain feedback where sensors measure the result of an action.

Engineering / workplace application: A pick-and-place system uses a controller, motor drives, joint actuators, position feedback, and a gripper as one coordinated system.

Illustrative example: Trace one pick cycle: controller command → drive → motor/joint motion → position sensor feedback → correction → gripper action.

Common mistake: Confusing a sensor with an actuator reverses the direction of information and can make a block diagram technically incorrect.

Module summary: Robotics is best understood as an integrated system whose characteristics and parts are selected for a real application.

OFFICIAL MODULE 2

Autonomous Robots: Line Follower and Obstacle Avoider

This module translates robotic principles into two basic autonomous systems. A line follower uses sensing and control to remain on a marked path; an obstacle avoider uses environmental sensing and movement decisions to avoid collisions. The official outline includes sensor assembly, motor-driver connection, microcontroller interfacing, and whole-system assembly.

Hours: 15

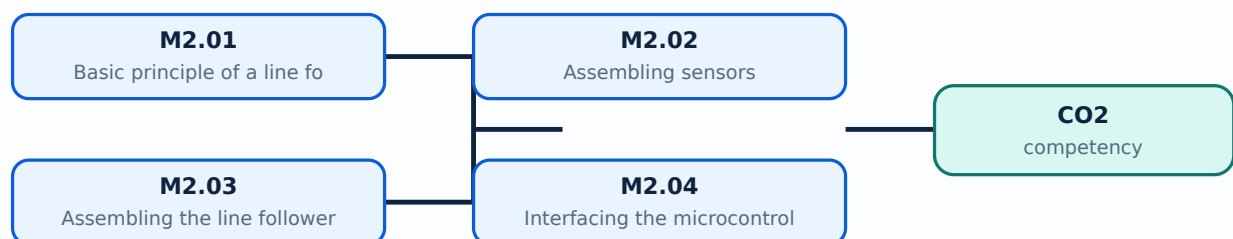
CO2 · Bloom level: Applying

Learning goals

- Explain colour sensing and the basic principle of a line follower.
- Assemble sensors, microcontroller, motor driver, and motors as a coherent system.
- Explain the basic movement and obstacle-avoidance principle and identify safe assembly and testing steps.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.01 — Basic principle of a line follower (2 hours)

Concept and explanation

A line follower detects the contrast between a track and its background, then adjusts left and right motor action to remain near the desired line. A sensor arrangement produces signals; the controller interprets them; the motor driver supplies controlled power to the motors. The response may be simple left/right correction or a proportional strategy. The design must define what sensor state corresponds to line position.

Malayalam support: Line follower contrast sensing രേഖ-സ്ഥിതി line-സ്ഥിതി position സ്ഥിതി left/right motor action സ്ഥിതി. Sensor → controller → motor driver → motors loop സ്ഥിതി സ്ഥിതി.

Study points

- Calibrate on both line and background; identify sensor polarity; test one motor at a time; define what happens when the line is lost; use a low-speed first test.

Engineering / workplace application: Line-following robots demonstrate feedback, closed-loop correction, sensor thresholds, and differential drive in a compact system.

Illustrative example: If the left sensor detects the line, the controller may reduce or stop the left motor and drive the right motor to correct position; the exact logic depends on calibration.

Common mistake: Assuming the sensor output is always active-high without checking the module can reverse the control logic.

Concept and explanation

Sensor assembly includes mechanical placement, electrical connection, power verification, signal identification, and calibration. Sensors must be mounted so that their field of view or sensing distance is appropriate. Wires should be secured and polarity checked before connecting a controller. Calibration records the sensor response on representative line, background, or obstacle conditions.

[illegible]

Study points

- Use a wiring table; confirm supply voltage and ground; keep sensor spacing symmetric when required; protect wires from moving parts; record raw or digital readings.

Engineering / workplace application: A calibrated pair of line sensors can provide left/right error information for a simple autonomous robot.

Illustrative example: Prepare a table with sensor name, VCC, GND, output, mounting height, and observed output on line and background.

Common mistake: Testing an unmounted or loosely mounted sensor and treating its noisy output as a control fault wastes time.

– M2.03 — Assembling the line follower (2 hours)

Concept and explanation

Line-follower assembly integrates the chassis, sensors, microcontroller, motor driver, motors, battery or regulated supply, and programme. Build in stages: inspect parts, mount sensors, connect common ground and supply, test sensor outputs, test motor-driver channels without a load, then perform low-speed track tests. Keep the centre of mass and sensor position suitable for the track.

Malayalam support: Line follower assembly step-by-step ഘട്ടങ്ങൾ: parts inspect, sensors mount, supply/ground connect, outputs test, motor driver test, ഘട്ടങ്ങൾ low-speed track test.

Study points

- Use a checklist and photographs or diagrams in the record; disconnect power while changing wiring; test each subsystem before integration; keep the battery secure.

Engineering / workplace application: The assembly demonstrates system integration and helps reveal whether a problem is mechanical, electrical, sensing, or control-related.

Illustrative example: Make a four-stage test record: sensor test, left/right motor test, combined bench test, and track test. Record the observed fault and corrective action.

Common mistake: Connecting the battery before checking polarity and common ground can damage the driver, controller, or sensors.

– M2.04 — Interfacing the microcontroller with motor driver and sensor IC (3 hours)

Concept and explanation

A microcontroller reads sensor signals and sends logic-level commands to a motor-driver interface. The driver handles the higher current needed by motors and may provide direction and enable inputs. A sensor IC or sensor board conditions the sensing signal. The interface requires compatible voltage levels, common reference, correct pin mapping, and a control programme that translates sensor states into motion.

Malayalam support: Microcontroller signal-level control ഏകദേശം; motor driver motor current handle വലിയ കറന്റ്. Common ground, pin mapping, voltage compatibility ഉറപ്പാക്കുന്നു.

Study points

- Draw separate signal and power paths; never power a motor directly from a microcontroller pin; check driver enable inputs; document pin numbers and active logic.

Engineering / workplace application: The same interface pattern appears in mobile robots, conveyor control, camera pan-tilt systems, and small automation cells.

Illustrative example: Create a pin map for two sensors, two motor directions, and two enable signals. Explain which signals are logic and which path carries motor power.

Common mistake: Using the controller supply as though it can deliver motor current causes resets, noise, or damage.

Concept and explanation

Robot movement results from the relationship among wheel arrangement, motor direction, speed, traction, and controller commands. In a differential-drive platform, changing the relative speeds of left and right wheels produces forward motion, turning, or rotation. Movement must be considered with friction, wheel slip, battery voltage, load, and obstacle contact. Autonomous movement is a repeated sense-decide-act cycle.

Malayalam support: Differential drive- Δ left/right wheel speed difference Δv forward motion, turn, rotation $\Delta \theta$. Movement- Δ traction, slip, load, battery voltage ΔV .

Study points

- Test forward, reverse, left, right, and stop separately; keep a clear emergency stop; limit speed during first trials; distinguish commanded motion from actual motion.

Engineering / workplace application: Mobile robots, AGVs, line followers, and obstacle avoiders all use controlled wheel motion to navigate a workspace.

Illustrative example: If the right wheel is faster than the left, the robot curves toward the slower side. Verify direction experimentally before writing control logic.

Common mistake: Assuming equal motor speed without calibration can cause a robot to drift even when both motors receive the same command.

– M2.06 — Assembling the obstacle avoider (3 hours)

Concept and explanation

An obstacle avoider uses a proximity or range sensor to detect an object and a decision rule to change motion. The official content includes sensor assembly, whole-system connection to the motor driver, and microcontroller interfacing. A safe procedure is to check the sensor field, define a threshold, test stationary readings, then test slow motion with a clear boundary and emergency stop.

Malayalam support: Obstacle avoider proximity/range sensor ഓബ്സ്റ്റാക്കിൾ ഓബ്സ്റ്റാക്കിൾ detect ദിറക്ഷൻ. Threshold, sensor field, slow test, emergency stop.

Study points

- Calibrate distance or detection threshold; ensure the sensor is not blocked by the chassis; define what happens when no obstacle is detected; test left and right turns independently.

Engineering / workplace application: Obstacle avoiders introduce reactive autonomy used in simple service robots, inspection platforms, and educational mobile systems.

Illustrative example: Write a decision table: clear path → move forward; obstacle near → stop; left side clear → turn left; otherwise turn right. Then identify one limitation of this rule.

Common mistake: A single threshold without considering sensor angle, surface reflectivity, or response delay may produce oscillation or late avoidance.

Concept and explanation

The official outline lists Series Test I under Module II. Revise the line follower principle, colour sensing, sensor assembly, whole-system assembly, microcontroller-motor-driver-sensor interface, basic movement, and obstacle avoider assembly. The test duration is listed as 1 hour.

Malayalam support: Series Test I- line follower, colour sensing, sensors, motor driver, microcontroller interface, movement, obstacle avoider പരീക്ഷ പരിശീലനം. Duration 1 hour പരീക്ഷ.

Study points

- Practise one system block diagram and one troubleshooting table; memorise the safe assembly sequence; use accurate component names.

Engineering / workplace application: A timed systems explanation tests whether the learner can connect individual components into a functioning autonomous robot.

Illustrative example: Draw a line-follower block diagram and annotate sensor input, controller decision, driver output, motor action, and feedback/correction.

Common mistake: Listing components without showing signal flow or power flow gives an incomplete answer.

Module summary: Autonomous robot construction depends on calibrated sensing, correct interfaces, safe power wiring, and a repeatable sense-decide-act cycle.

OFFICIAL MODULE 3

Sensors in Robotic Applications

Sensors provide information about a robot's internal state and its external environment. Selection is an engineering decision based on the variable to measure, range, resolution, response, noise, interface, mounting, cost, and

environmental conditions. This module covers internal, external, miscellaneous sensors, and variable-recognition systems.

Hours: 15

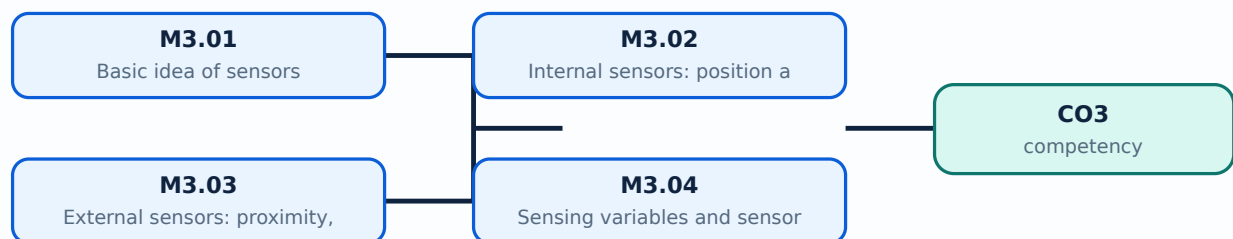
CO3 · Bloom level: Applying

Learning goals

- Explain the basic idea of sensing and distinguish internal from external sensors.
- Compare position, velocity, proximity, range, touch/slip, force/torque, and other sensors.
- Select sensors for variables such as smell, temperature, humidity, light, speech, and voice recognition.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.01 — Basic idea of sensors (1 hours)

Concept and explanation

A sensor detects a physical or environmental variable and produces a usable signal. The signal may be analogue or digital, and may be processed by a controller before a decision is made. A sensor specification should identify the measurand, range, sensitivity, resolution, accuracy, response time, interface, mounting, and operating environment.

Malayalam support: Sensor ഏതു physical variable detect ചെയ്ത് ഉപയോഗ്യമായ usable signal ഉത്പാദിപ്പിക്കുന്നു. Measurand, range, sensitivity, resolution, accuracy, response time, interface specification-യെ പരിശോധിക്കുന്നു.

Study points

- Start with the variable and required decision; distinguish sensor output from interpreted information; check calibration and environmental limits; document units.

Engineering / workplace application: A robot may use joint encoders for internal state and proximity sensors for external interaction.

Illustrative example: For a robot that must stop before a wall, state the measurand, approximate range, required response, and one environmental issue before naming a sensor.

Common mistake: Choosing a sensor by name alone, without specifying range or environment, is not a defensible selection.

– M3.02 — Internal sensors: position and velocity sensors (3 hours)

Concept and explanation

Internal sensors measure variables inside the robot, especially joint or wheel position and velocity. Position sensing may use optical or non-optical methods; velocity may be measured directly or estimated from position change over time. Selection depends on resolution, speed, backlash, mounting, feedback requirement, and environmental exposure. The sensor must be connected to the controller in a way that supports stable feedback.

Malayalam support: Internal sensor robot-നുള്ള അവസ്ഥാ അളവ് അളക്കുന്നു.

Position sensor joint/wheel position അളക്കുന്നു; velocity direct നോട്ട് അളക്കുന്നു position change നോട്ട് estimate അളക്കുന്നു.

Study points

- Separate absolute and relative position ideas; consider zero reference and calibration; choose resolution according to required movement; account for noise when differentiating position for velocity.

Engineering / workplace application: Joint feedback supports trajectory control, while wheel encoders support odometry and speed regulation in mobile robots.

Illustrative example: If one wheel encoder gives 400 counts per revolution and wheel circumference is 200 mm, ideal distance per count is 0.5 mm before slip and calibration effects.

Common mistake: Treating an encoder count as a physical distance without wheel radius, gearing, or calibration leads to incorrect motion estimates.

– M3.03 — External sensors: proximity, force and torque (3 hours)

Concept and explanation

External sensors measure the robot's relationship with its environment. Proximity sensors detect nearby objects; contact and non-contact sensing differ in how interaction occurs. Range sensing estimates distance. Touch and slip sensors help detect grasp or object motion. Force and torque sensors measure interaction loads and support compliant or guarded operation. Selection depends on surface, range, speed, contact, and safety needs.

Malayalam support: External sensor robot-റോബറ്റ് environment-പരിസ്ഥിതി relationship measure അളക്കുന്നു. Proximity, range, touch/slip, force/torque sensing-വിവിധ different use cases-ഉപയോഗങ്ങൾ.

Study points

- Define whether contact is allowed; consider object material and orientation; distinguish detection from measurement; include overload and safe-failure behaviour for force sensing.

Engineering / workplace application: A gripper can use touch or force feedback to avoid crushing an object; a mobile robot can use range sensing for obstacle avoidance.

Illustrative example: Select one sensor for detecting a nearby wall and another for measuring grip force. Compare range, contact, output, and safety considerations.

Common mistake: A proximity sensor may indicate that something is near but not provide a reliable dimension or identity without further processing.

– M3.04 — Sensing variables and sensors for robotics (4 hours)

Concept and explanation

Robotic systems may sense position, velocity, acceleration, distance, touch, force, torque, temperature, humidity, light, sound, smell, and other variables. The syllabus names optical, inertial, thermal, chemical, biosensor, and other common sensing contexts. Selection is guided by the variable, physical principle, operating range, noise, response, environment, and controller interface.

Malayalam support: Robot sensing variables ഏകദേശം: position, velocity, acceleration, distance, touch, force, temperature, humidity, light, sound, smell. Variable ഏകദേശം sensor select ചെയ്യേണ്ട.

Study points

- Prepare a variable-sensor-application table; state one limitation for every selected sensor; check whether the output needs amplification, filtering, or conversion.

Engineering / workplace application: Thermal sensors support process monitoring; optical sensors support line tracking; inertial sensors support orientation estimation; chemical or biosensors support specialised detection.

Illustrative example: Build a table with variable, possible sensor family, output type, and one robotic application for temperature, light, distance, and acceleration.

Common mistake: Saying “use a sensor” without naming the variable and operating condition does not complete the engineering reasoning.

– M3.05 — Variable-recognition systems (4 hours)

Concept and explanation

Recognition systems interpret sensor data to identify a state, object, voice, or event. The sensing element alone does not perform recognition; signal conditioning, feature extraction, thresholds or models, and decision logic may be required. The syllabus specifically includes speech or voice recognition systems and also lists smell, heat/temperature, humidity, and light as sensing variables.

Malayalam support: Recognition system sensor reading-സ്ഥിതാവസ്ഥ/വസ്തു/വോയ്സ് തിരിച്ചറിയുന്നു. Sensor തിരിച്ചറിയുന്നു; processing, features, threshold/model, decision logic തിരിച്ചറിയുന്നു.

Study points

- Distinguish raw measurement from classification; define false positive and false negative risks; test with representative conditions; provide a fallback when confidence is low.

Engineering / workplace application: Voice interfaces, object sorting, environmental monitoring, and self-driving functions combine multiple sensor readings with recognition logic.

Illustrative example: For voice recognition in a noisy workshop, list two preprocessing or decision safeguards and explain why a fallback command is needed.

Common mistake: Assuming one noisy reading is enough for recognition can cause unstable decisions and unsafe robot behaviour.

Module summary: Good sensor selection begins with the variable and task, then weighs range, accuracy, response, interface, environment, and safety.

OFFICIAL MODULE 4

Actuators, Drives and Sensor Selection

Actuators convert energy into controlled physical action. A robot's building blocks, motor type, drive, transmission, control method, and sensors must be

selected together. The module also introduces nontraditional sensors and actuators and applies selection to maze-solving robots and self-driving cars.

Hours: 15

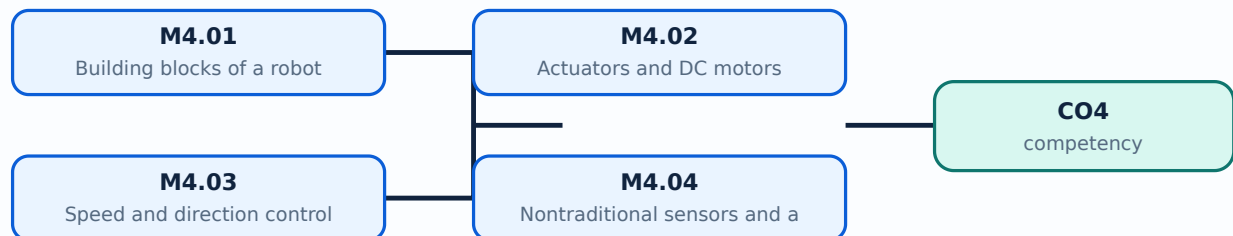
CO4 · Bloom level: Applying

Learning goals

- Identify the building blocks of a robot and the role of actuators and drives.
- Compare DC, servo, stepper, direct, and nontraditional actuation options.
- Select sensors and actuators for speed/direction control, obstacle avoidance, navigation, path planning, maze solving, and self-driving cars.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

– M4.01 — Building blocks of a robot (2 hours)

Concept and explanation

A robot can be represented as a chain of sensing, control, actuation, mechanical transmission, structure, power, and end-effector or mobility elements. The controller receives information, applies a decision or control law, and commands a drive. The actuator produces motion or force, while the mechanical structure transfers it to the task. Feedback closes the loop when the result is measured.

Malayalam support: Robot building blocks: sensing, control, drive, actuator, transmission, structure, power, end effector/mobility. Feedback ഏകദേശം result measure ഏകദേശം control ഏകദേശം.

Study points

- Draw separate information and energy paths; identify the controlled variable; include the environment and feedback sensor; do not omit the power source.

Engineering / workplace application: This block view helps troubleshoot whether a robot fault originates in sensing, logic, power, drive, mechanics, or the task interface.

Illustrative example: Create a block diagram for a mobile robot: battery, controller, sensor, motor driver, DC motors, wheels, chassis, and feedback path.

Common mistake: Drawing only the controller and motors hides the sensing and mechanical elements that determine actual robot behaviour.

— M4.02 — Actuators and DC motors (3 hours)

Concept and explanation

An actuator creates a controlled physical output such as motion or force. The syllabus identifies DC, servo, and stepper motors as electric motor types used in robotic actuation. A DC motor is suited to continuous rotation and speed control; a servo system combines a motor with feedback and control for commanded position or motion; a stepper motor moves in discrete steps and can be useful where controlled incremental motion is required. Selection must consider torque, speed, duty, load, resolution, power, and feedback.

Malayalam support: Actuator physical action ഏകദേശം. DC continuous rotation; servo feedback-based position/motion control; stepper discrete steps. Torque, speed, load, duty, power, feedback പരിഗണിക്കുക.

Study points

- Separate motor type from complete drive system; calculate or estimate load requirements before selection; check stall current and heat; use feedback when open-loop motion is not reliable.

Engineering / workplace application: Wheels, joints, grippers, pan-tilt units, and conveyor mechanisms use different actuator choices.

Illustrative example: For a gripper, compare a servo and a DC motor with encoder. Explain which provides simpler position control and what trade-off remains.

Common mistake: Selecting a motor only by no-load speed ignores torque, load, acceleration, gearbox, and stall-current requirements.

– M4.03 — Speed and direction control of drives (3 hours)

Concept and explanation

Motor drives regulate the energy delivered to actuators. Speed may be controlled by a command such as PWM in a suitable driver, while direction may be changed through polarity or an H-bridge arrangement for compatible DC motors. The drive must be matched to motor voltage, current, switching, protection, controller logic, and thermal limits. Closed-loop speed control uses feedback to correct load-related variation.

Malayalam support: Drive motor-എന്ന ഊർജ്ജ നിയന്ത്രണം. DC മോട്ടോർ സ്പീഡ് PWM നിയന്ത്രണം method-എന്ന ഏകദേശമായി; direction H-bridge/polarity എന്നിവ നിയന്ത്രണം. Voltage/current/protection match നിയന്ത്രണം.

Study points

- Check driver voltage and continuous/peak current; provide common reference and flyback or built-in protection; test direction with wheels lifted; limit duty cycle during first tests.

Engineering / workplace application: Differential-drive robots use independent left/right speed control to move, turn, and rotate.

Illustrative example: Prepare a direction table for a two-wheel robot: both forward, both reverse, left slower, right slower, and one wheel stopped. State the resulting motion.

Common mistake: Connecting a motor directly to a microcontroller output or reversing direction without a suitable driver can damage the system.

Concept and explanation

The outline distinguishes common electric motors from nontraditional sensors and actuators. Nontraditional devices may use special materials, fluid power, smart-material effects, or unusual sensing principles. Selection is based on the required output, response, force, stroke, repeatability, energy source, environment, and control interface. The term should be explained with the chosen example rather than treated as a single device.

Malayalam support: Nontraditional sensor/actuator ഏതെങ്കിലും conventional device-ഏതെങ്കിലും sensing/actuation principle ഏതെങ്കിലും device നൽകുന്നു. Output, response, stroke, energy source, environment ഏതെങ്കിലും ഉൾപ്പെടുന്നു.

Study points

- State the physical principle; identify the input and output; compare response and controllability with a conventional alternative; include limitations and safety.

Engineering / workplace application: Special actuators can be useful in miniature systems, soft robotics, precision positioning, or environments where conventional motors are unsuitable.

Illustrative example: Choose one smart-material or fluidic actuator example and describe its input, output, useful feature, and one limitation.

Common mistake: Using “nontraditional” as a vague label without naming a principle or application makes the explanation incomplete.

– M4.05 — Optical, inertial, thermal, chemical, biosensor and other common sensors (3 hours)

Concept and explanation

The syllabus lists optical, inertial, thermal, chemical, biosensor, and other common sensors in the context of actuator and sensor selection. Optical sensing can support line, distance, or vision tasks; inertial sensing supports acceleration and orientation; thermal sensing measures temperature; chemical and biosensors detect substances or biological conditions. The correct choice depends on the variable, range, environment, response, calibration, and decision required.

Malayalam support: Optical, inertial, thermal, chemical, biosensor ഏകദേശം different variables measure ചെയ്യുന്നു. Application-ൽ ഏകദേശം variable, range, response, calibration ഏകദേശം select ചെയ്യുന്നു.

Study points

- Build a selection matrix; state whether the sensor is internal or external; mention environmental interference; explain how its output reaches the controller.

Engineering / workplace application: A self-driving car combines optical/range, inertial, wheel, and other sensors because no single sensor covers every navigation and safety need.

Illustrative example: For a self-driving-car lane and temperature-monitoring task, choose two sensor families and explain why one cannot replace the other.

Common mistake: Treating different sensor families as interchangeable ignores their measurand and operating limitations.

– M4.06 — Selection of sensors and actuators for self-driving cars (2 hours)

Concept and explanation

A self-driving car requires coordinated localisation, navigation, obstacle avoidance, and path planning. Sensor selection may include optical or range sensing for surroundings, inertial sensing for motion, wheel or actuator feedback, and other sensors for vehicle state. Actuator selection must consider steering, propulsion, braking, speed, direction, response, redundancy, and fail-safe behaviour. A choice should be justified by function and environment, not by a component list alone.

Malayalam support: Self-driving car-□ localisation, navigation, obstacle avoidance, path planning □□□□□□□□□□ multiple sensors □□□□. Steering, propulsion, braking actuators-□□ fail-safe behaviour □□□□□□□□□□.

Study points

- Map each function to a sensor/actuator; identify redundancy or fallback; consider weather, occlusion, latency, calibration, power, and safety; separate prototype choice from production qualification.

Engineering / workplace application: The same selection reasoning applies to autonomous delivery vehicles, warehouse AGVs, and maze-solving robots.

Illustrative example: Prepare a table with function, sensor, actuator, decision output, and one failure response for lane following, obstacle avoidance, speed control, and emergency stop.

Common mistake: Assuming one camera or one distance sensor can guarantee safe autonomy ignores occlusion, uncertainty, and failure modes.

– M4.ST2 — Series Test II (1 hours)

Concept and explanation

The official outline lists Series Test II under Module IV. Revise robot building blocks, motor types, drives, speed and direction control, actuator selection, nontraditional sensors/actuators, optical/inertial/thermal/chemical/biosensors, and sensor-actuator selection for maze-solving robots and self-driving cars. The test duration is listed as 1 hour.

Malayalam support: Series Test II-രണ്ടാം ഘട്ടം building blocks, motor types, drives, control, sensor/actuator selection, self-driving case study പരിശീലനം revise പരീക്ഷണം. Duration 1 hour 60.

Study points

- Practise one selection matrix and one labelled robot block diagram; state the reason for every component choice; include safety and failure response.

Engineering / workplace application: Case-study revision helps connect definitions with engineering selection decisions.

Illustrative example: For a maze-solving robot, select one proximity sensor and one actuator, then write the condition under which each choice would be changed.

Common mistake: Listing a sensor or motor without stating the function it serves produces an unsupported design answer.

Module summary: Robot design is a systems-selection problem: sensing, actuation, drives, control, structure, power, and safety must agree with the application.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

[Open the module to view its labelled learning-map diagram.](#)

[Module 2: Autonomous](#)

[Open the module to view its labelled learning-map diagram.](#)

[Module 3: Sensors in](#)

[Open the module to view its labelled learning-map diagram.](#)

[Module 4: Actuators, Drives](#)

[Open the module to view its labelled learning-map diagram.](#)

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied PDF does not provide a separate self-learning activity list. Treat the line-follower, obstacle-avoider, sensor-selection, actuator-selection, and self-driving-car case-study items in the official modules as guided learning tasks.

Practical requirements / equipment

- No separate equipment list is supplied in the PDF. Robot assembly and drive experiments require departmental approval, suitable low-voltage components, current-limited power, and supervision.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- The supplied PDF lists a Series Test component of 2 hours in total.
- Series Test I is listed under Module II for 1 hour.
- Series Test II is listed under Module IV for 1 hour.
- The PDF does not specify CIE/ESE marks or a separate end-semester examination pattern; this lesson does not invent one.

Module-wise Questions and Answers

module-1 — Fundamentals and Applications of Robots

1. Explain Fundamentals, types and need of robots. [3 marks]

A robot is a programmable machine capable of carrying out actions through a combination of structure, actuation, sensing, and control. Robotics is the field concerned with the design, construction, operation, and application of robots. The need for robots arises when tasks are repetitive, hazardous, inaccessible, highly precise, or require sustained operation. Types may be classified by application, mobility, control, mechanical configuration, or level of autonomy; the classification should state the basis used.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Characteristics of robots. [3 marks]

Important characteristics include programmability, repeatability, accuracy, payload, reach or workspace, degrees of freedom, speed, resolution, reliability, and the ability to sense or respond to conditions. No single characteristic is universally best: a high-speed robot may not be the most accurate, and a large workspace may require a heavier structure. Characteristics must be related to the application and specified conditions.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Applications of robots. [3 marks]

The syllabus lists medical, mining, space, defence, security, domestic, entertainment, and industrial applications. Applications differ in environment, human interaction, payload, sensing, autonomy, safety, and reliability. A useful application analysis identifies the task, environment, human risk, required sensors, actuator demands, and the reason a robot is preferable to direct human operation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Industrial applications. [3 marks]

Industrial robot applications in the syllabus include material handling, welding, spray painting, and machining. Material handling depends on payload, reach, gripper design, and cycle time. Welding needs path accuracy and process coordination. Spray painting needs controlled motion, coverage, and safe handling of vapour or overspray. Machining needs stiffness, tool control, and suitable force management. Application-specific tooling is often as important as the robot arm.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Autonomous Robots: Line Follower and Obstacle Avoider

5. Explain Basic principle of a line follower. [3 marks]

A line follower detects the contrast between a track and its background, then adjusts left and right motor action to remain near the desired line. A sensor arrangement produces signals; the controller interprets them; the motor driver supplies controlled power to the motors. The response may be simple left/right correction or a proportional strategy. The design must define what sensor state corresponds to line position.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Assembling sensors. [3 marks]

Sensor assembly includes mechanical placement, electrical connection, power verification, signal identification, and calibration. Sensors must be mounted so that their field of view or sensing distance is appropriate. Wires should be secured and polarity checked before connecting a controller. Calibration records the sensor response on representative line, background, or obstacle conditions.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Assembling the line follower. [3 marks]

Line-follower assembly integrates the chassis, sensors, microcontroller, motor driver, motors, battery or regulated supply, and programme. Build in stages: inspect parts, mount sensors, connect common ground and supply, test sensor outputs, test motor-driver channels without a load, then perform low-speed track tests. Keep the centre of mass and sensor position suitable for the track.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Interfacing the microcontroller with motor driver and sensor IC. [3 marks]

A microcontroller reads sensor signals and sends logic-level commands to a motor-driver interface. The driver handles the higher current needed by motors and may provide direction and enable inputs. A sensor IC or sensor board conditions the sensing signal. The interface requires compatible voltage levels, common reference, correct pin mapping, and a control programme that translates sensor states into motion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Sensors in Robotic Applications

9. Explain Basic idea of sensors. [3 marks]

A sensor detects a physical or environmental variable and produces a usable signal. The signal may be analogue or digital, and may be processed by a controller before a decision is made. A sensor specification should identify the measurand, range, sensitivity, resolution, accuracy, response time, interface, mounting, and operating environment.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Internal sensors: position and velocity sensors. [3 marks]

Internal sensors measure variables inside the robot, especially joint or wheel position and velocity. Position sensing may use optical or non-optical methods; velocity may be measured directly or estimated from position change over time. Selection depends on resolution, speed,

backlash, mounting, feedback requirement, and environmental exposure. The sensor must be connected to the controller in a way that supports stable feedback.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain External sensors: proximity, force and torque. [3 marks]

External sensors measure the robot's relationship with its environment. Proximity sensors detect nearby objects; contact and non-contact sensing differ in how interaction occurs. Range sensing estimates distance. Touch and slip sensors help detect grasp or object motion. Force and torque sensors measure interaction loads and support compliant or guarded operation. Selection depends on surface, range, speed, contact, and safety needs.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Sensing variables and sensors for robotics. [3 marks]

Robotic systems may sense position, velocity, acceleration, distance, touch, force, torque, temperature, humidity, light, sound, smell, and other variables. The syllabus names optical, inertial, thermal, chemical, biosensor, and other common sensing contexts. Selection is guided by the variable, physical principle, operating range, noise, response, environment, and controller interface.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Actuators, Drives and Sensor Selection

13. Explain Building blocks of a robot. [3 marks]

A robot can be represented as a chain of sensing, control, actuation, mechanical transmission, structure, power, and end-effector or mobility elements. The controller receives information, applies a decision or control law, and commands a drive. The actuator produces motion or force, while the mechanical structure transfers it to the task. Feedback closes the loop when the result is measured.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Actuators and DC motors. [3 marks]

An actuator creates a controlled physical output such as motion or force. The syllabus identifies DC, servo, and stepper motors as electric motor types used in robotic actuation. A DC motor is suited to continuous rotation and speed control; a servo system combines a motor with feedback and control for commanded position or motion; a stepper motor moves in discrete steps and can be useful where controlled incremental motion is required. Selection must consider torque, speed, duty, load, resolution, power, and feedback.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Speed and direction control of drives. [3 marks]

Motor drives regulate the energy delivered to actuators. Speed may be controlled by a command such as PWM in a suitable driver, while direction may be changed through polarity or an H-bridge arrangement for compatible DC motors. The drive must be matched to motor voltage, current, switching, protection, controller logic, and thermal limits. Closed-loop speed control uses feedback to correct load-related variation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Nontraditional sensors and actuators. [3 marks]

The outline distinguishes common electric motors from nontraditional sensors and actuators. Nontraditional devices may use special materials, fluid power, smart-material effects, or unusual sensing principles. Selection is based on the required output, response, force, stroke, repeatability, energy source, environment, and control interface. The term should be explained with the chosen example rather than treated as a single device.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Fundamentals, types and need of robots* with one relevant application. (5 marks)
2. **2.** Define or explain *Characteristics of robots* with one relevant application. (5 marks)
3. **3.** Define or explain *Basic principle of a line follower* with one relevant application. (5 marks)
4. **4.** Define or explain *Assembling sensors* with one relevant application. (5 marks)
5. **5.** Define or explain *Basic idea of sensors* with one relevant application. (5 marks)
6. **6.** Define or explain *Internal sensors: position and velocity sensors* with one relevant application. (5 marks)
7. **7.** Define or explain *Building blocks of a robot* with one relevant application. (5 marks)
8. **8.** Define or explain *Actuators and DC motors* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Fundamentals and Applications of Robots:** Robotics is best understood as an integrated system whose characteristics and parts are selected for a real application.
- **Autonomous Robots: Line Follower and Obstacle Avoider:** Autonomous robot construction depends on calibrated sensing, correct interfaces, safe power wiring, and a repeatable sense-decide-act cycle.
- **Sensors in Robotic Applications:** Good sensor selection begins with the variable and task, then weighs range, accuracy, response, interface, environment, and safety.

- **Actuators, Drives and Sensor Selection:** Robot design is a systems-selection problem: sensing, actuation, drives, control, structure, power, and safety must agree with the application.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Fundamentals, types and need of robots	A robot is a programmable machine capable of carrying out actions through a combination of structure, actuation, sensing, and control. Robotics is the field concerned with the design, construction, operation, and application of robots. The need for robots arises
Characteristics of robots	Important characteristics include programmability, repeatability, accuracy, payload, reach or workspace, degrees of freedom, speed, resolution, reliability, and the ability to sense or respond to conditions. No single characteristic is universally best: a high
Applications of robots	The syllabus lists medical, mining, space, defence, security, domestic, entertainment, and industrial applications. Applications differ in environment, human interaction, payload, sensing, autonomy, safety, and reliability. A useful application analysis identifies
Industrial applications	Industrial robot applications in the syllabus include material handling, welding, spray painting, and machining. Material handling depends on payload, reach, gripper design, and cycle time. Welding needs path accuracy and process coordination. Spray painting needs
Parts and functions of robots	A robot system commonly includes a manipulator or mechanical structure, joints and links, actuators, drives, sensors, controller, power supply, end effector, and user/programming interface. The mechanical structure positions the tool, actuators produce motion,

Term	Short meaning
Basic principle of a line follower	A line follower detects the contrast between a track and its background, then adjusts left and right motor action to remain near the desired line. A sensor arrangement produces signals; the controller interprets them; the motor driver supplies controlled power
Assembling sensors	Sensor assembly includes mechanical placement, electrical connection, power verification, signal identification, and calibration. Sensors must be mounted so that their field of view or sensing distance is appropriate. Wires should be secured and polarity check
Assembling the line follower	Line-follower assembly integrates the chassis, sensors, microcontroller, motor driver, motors, battery or regulated supply, and programme. Build in stages: inspect parts, mount sensors, connect common ground and supply, test sensor outputs, test motor-driver c
Interfacing the microcontroller with motor driver and sensor IC	A microcontroller reads sensor signals and sends logic-level commands to a motor-driver interface. The driver handles the higher current needed by motors and may provide direction and enable inputs. A sensor IC or sensor board conditions the sensing signal. Th
Basic principle of moving	Robot movement results from the relationship among wheel arrangement, motor direction, speed, traction, and controller commands. In a differential-drive platform, changing the relative speeds of left and right wheels produces forward motion, turning, or rotati
Assembling the obstacle avoider	An obstacle avoider uses a proximity or range sensor to detect an object and a decision rule to change motion. The official content includes sensor assembly, whole-system connection to the motor driver, and microcontroller interfacing. A safe procedure is to c
Series Test I	The official outline lists Series Test I under Module II. Revise the line follower principle, colour sensing, sensor assembly, whole-system assembly, microcontroller-motor-driver-sensor interface, basic movement, and obstacle avoider assembly. The test duratio
Basic idea of sensors	A sensor detects a physical or environmental variable and produces a usable signal. The signal may be analogue or digital, and may be processed by a controller before a decision is made. A sensor specification should identify the measurand, range, sensitivity,
Internal sensors: position and velocity sensors	Internal sensors measure variables inside the robot, especially joint or wheel position and velocity. Position sensing may use optical or non-optical methods; velocity may be measured directly or estimated from position change over time. Selection depends on r
External sensors: proximity, force and torque	External sensors measure the robot's relationship with its environment. Proximity sensors detect nearby objects; contact and non-contact sensing differ in how interaction occurs. Range sensing estimates distance. Touch and slip sensors help detect grasp or obj
Sensing variables and sensors for robotics	Robotic systems may sense position, velocity, acceleration, distance, touch, force, torque, temperature, humidity, light, sound, smell, and other variables. The syllabus names optical, inertial, thermal, chemical, biosensor, and other common sensing contexts.

Term	Short meaning
Variable-recognition systems	Recognition systems interpret sensor data to identify a state, object, voice, or event. The sensing element alone does not perform recognition; signal conditioning, feature extraction, thresholds or models, and decision logic may be required. The syllabus spec
Building blocks of a robot	A robot can be represented as a chain of sensing, control, actuation, mechanical transmission, structure, power, and end-effector or mobility elements. The controller receives information, applies a decision or control law, and commands a drive. The actuator p
Actuators and DC motors	An actuator creates a controlled physical output such as motion or force. The syllabus identifies DC, servo, and stepper motors as electric motor types used in robotic actuation. A DC motor is suited to continuous rotation and speed control; a servo system com
Speed and direction control of drives	Motor drives regulate the energy delivered to actuators. Speed may be controlled by a command such as PWM in a suitable driver, while direction may be changed through polarity or an H-bridge arrangement for compatible DC motors. The drive must be matched to mo
Nontraditional sensors and actuators	The outline distinguishes common electric motors from nontraditional sensors and actuators. Nontraditional devices may use special materials, fluid power, smart-material effects, or unusual sensing principles. Selection is based on the required output, respons
Optical, inertial, thermal, chemical, biosensor and other common sensors	The syllabus lists optical, inertial, thermal, chemical, biosensor, and other common sensors in the context of actuator and sensor selection. Optical sensing can support line, distance, or vision tasks; inertial sensing supports acceleration and orientation; t
Selection of sensors and actuators for self-driving cars	A self-driving car requires coordinated localisation, navigation, obstacle avoidance, and path planning. Sensor selection may include optical or range sensing for surroundings, inertial sensing for motion, wheel or actuator feedback, and other sensors for vehi
Series Test II	The official outline lists Series Test II under Module IV. Revise robot building blocks, motor types, drives, speed and direction control, actuator selection, nontraditional sensors/actuators, optical/inertial/thermal/chemical/biosensors, and sensor-actuator s

Official References and Resources

References listed in the supplied syllabus

1. Koren Yoram, Robotics for Engineers, McGraw-Hill Education, New Delhi, 1st Edition.
2. Hedge, G. S., Textbook on Industrial Robotics, Laxmi Publications, New Delhi, 1st Edition.
3. Groover, Mikell P., Industrial Robotics: Technology, Programming and Applications, McGraw-Hill Education, New Delhi, 2nd Edition.

4. K. K. Appu Kuttan, Robotics, I. K. International, 2007.
5. Fu, K. S.; Gonzalez, R. C.; Lee, C. S. G., Robotics, McGraw-Hill Education, New Delhi Pvt. Ltd.
6. Sawney, A. K. and Puneet Sawney, A Course in Mechanical Measurements and Instrumentation and Control, 12th edition, Dhanpat Rai & Co., New Delhi, 2013.

Official learning resources listed

- <https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-to-robotics-fall-2005/index.html>
- <https://nptel.ac.in/courses/112/101/112101098/>
- <https://nptel.ac.in/courses/112/105/112105249/>
- <https://create.arduino.cc/projecthub/robocircuits/line-follower-robot-arduino-299bae>
- <https://create.arduino.cc/projecthub/Alfa0420/obstacle-avoiding-robot-using-arduino-and-ultrasonic-sensor-c179bb>

In-page Search

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